Qn 2	Answer	Marks
(a)(i)	$pV = nRT \quad \Rightarrow \quad n = \frac{pV}{RT} = \frac{(4.0\times10^5)(1.2\times10^{-3})}{8.31(290)}$ $= 0.199\,mol = 0.20\,mol$	[1] [1]
(a)(ii)	Consider C to A, volume is constant so $\frac{p}{T} = constant$ $\frac{p_c}{T_c} = \frac{p_A}{T_A}$ $T_c = \frac{p_c}{p_A}T_A = \frac{(7.75\times10^5)(290)}{4.0\times10^5}$ $= 561.9\ K = 562\ K$ (accepted range of 558 K to 566 K) Accepted range of $P_c = (7.75 \pm 0.05)\times10^5$ Pa	[1] [1]
(a)(iii)	(This is an isovolumetric change) Work done = area under p-V graph = 0 J	[1]
(b)	During one complete cycle ABCA, increase in internal energy is zero. ($\Delta U=0$) Net work done on the gas = area enclosed by ABCA = positive value, as volume contracts during B to C Applying first law of thermodynamics, the increase in internal energy (ΔU) of the system is equal to the sum of heat supplied (Q) and work done on (W) the system. $\Delta U= Q + W$ $Q=\Delta U - W = 0 - (+\text{ value}) = -\text{ value}$ Heat removed from gas during one cycle.	[1] [1] [1]
(c)	$pV = nRT$ Since $V, R\ and\ T\ constant,\ p \propto n$ $loss = \frac{4}{100}\times 0.199 = 0.00796\,mol$ $= 0.00796\times6.02\times10^{23} = 4.79192\times10^{21}\ atoms$ $rate = \frac{(4.792\times10^{21})}{3\times24\times60\times60}$ $= 1.85\times10^{16}\ s^{-1}$	[1] [1] [1]
	Max Marks	**11**

Qn 3	Answer	Marks